Efficacy and Safety of Pulmora in Moderate-to-severe asthma: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Pulmora (PLM) is an inhaled long-acting biologic-adjunct under investigation for moderate-to-severe asthma. **Methods:** In this multicentre trial, 731 adults were randomised 1:1 to Pulmora or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was annual exacerbation rate. **Results:** The primary endpoint was 0.62 with Pulmora versus 1.04 with placebo (p 0.003). Nasopharyngitis occurred in 14.8% of Pulmora patients and 12.2% of placebo patients. **Conclusions:** Pulmora met its primary endpoint with a manageable safety profile in moderate-to-severe asthma.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of moderate-to-severe asthma and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Pulmora	Placebo
Participants (n)	365	366
Mean age (years)	55.3	58.6
Female (%)	55	54
Primary endpoint	0.62	1.04
p-value	0.003	-
Nasopharyngitis (%)	14.8	12.2
Discontinuation due to AE (%)	4.2	3.0

Serious adverse events were reported in 8.7% of the Pulmora group and 5.6% of the placebo group. There were 4 deaths in the Pulmora arm and 3 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Pulmora is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.